PHYS 357: Honours Quantum Physics 1

W.R.YCH.

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Abstract

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Contents

1 Introduction

Textbook used is JS Townsend, *A Modern Approach to Quantum Mechanics*, 2nd ed., University Science Books 2012. Taught by Bill Coish. Mondays, Wednesdays and Fridays from 9:35 to 10:25 in Steward Biology Building N2/2. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on Wednesdays, best 8 of 12).
- 20% midterms (best of 2, October 2nd and November 6th).
- 30% final exam.

2 Stern-Gerlach Experiments

2.1 The classical picture

2.1.1 Angular momentum

Definition 1 (Orbital angular momentum)**.** *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \tag{1}$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x-y plane, only the z component survives. On a circular path of radius r with speed v,

$$L_z = mvr.$$

Definition 2 (Angular momentum of a rigid body)**.** *For a rigid body*[1] *rotating about a fixed axis this is*

$$L = I_0\omega, \tag{2}$$

with I_0 the moment of inertia and ω the angular velocity.

Remark 1. *The moment of inertia I_0 measures how mass is distributed relative to the rotation axis.*

2.1.2 Magnetic moment of a circulating charge

Definition 3 (Magnetic moment)**.** *A magnetic moment is a measure of the net circulation of electric charge. It is a vector quantity that determines the torque a magnet will experience in an external magnetic field.*

[1] A rigid body is one that does not deform under the action of external forces.

The magnetic moment of a point particle of mass m and charge q in a circular orbit of radius r and period T is

$$\mu = \frac{IA}{c}[2] = \frac{1}{c}\left(\frac{qv}{2\pi r}\right)\left(\pi r^2\right) = \frac{q}{2mc}\,mvr = \frac{q}{2mc}\,L_z, \tag{3}$$

where I is the current and A is the area enclosed by the orbit. A single circulating charge counts as a steady current $I = q/T = qv/2\pi r$ because it passes any given point on the loop once per period.

2.1.3 Gyromagnetic ratio

Definition 4 (Gyromagnetic ratio)**.** *The gyromagnetic ratio γ is the constant of proportionality between the magnetic moment and the angular momentum,*

$$\vec{\mu} = \gamma\vec{L}. \tag{4}$$

For the circulating charge above,

$$\gamma = \frac{q}{2mc}. \tag{5}$$

Remark 2. *This is a purely classical result, since the charge is treated as a point particle on a definite orbit with a definite radius and speed, and nothing is quantized. It matters here because measuring $\vec{\mu}$ is how we measure $\vec{L}$, and it is what the Stern-Gerlach apparatus exploits.*

2.2 Spin

Definition 5 (Spin)**.** *Spin $\vec{S}$ is an intrinsic angular momentum carried by a particle. It is a fixed property of the particle type, like charge or mass, and does not come from any motion through space. A stationary electron alone in empty space still has spin.*

In quantum mechanics the spin $\vec{S}$ replaces the orbital angular momentum $\vec{L}$ of the classical picture.

2.2.1 Intrinsic moment and the g-factor

The magnetic moment is still proportional to the angular momentum, but the gyromagnetic ratio picks up an extra dimensionless factor,

$$\vec{\mu} = \gamma\vec{S}, \qquad \gamma = \frac{qg}{2mc}, \tag{6}$$

where g is the g-factor of the particle.

Definition 6 (g-factor)**.** *The g-factor of a particle*[3] *is the dimensionless number relating its magnetic moment to its spin angular momentum. It measures how much stronger or weaker the moment is than the classical value $\gamma = q/2mc$, which corresponds to $g = 1$.*

[2] Here c is in Gaussian units.

[3] Its value must be measured or derived from relativistic quantum theory.

2.2.2 Magnetons

Definition 7 (Planck constant)**.** *The Planck constant h is the fundamental constant of quantum mechanics. It sets the scale at which quantization matters, and relates the energy of a photon to its frequency by $E = h\nu$. Its value is*

$$h \approx 6.626 \times 10^{-34} \text{ J s}. \tag{7}$$

It has units of $J \cdot s$, which is the same as angular momentum.

Definition 8 (Reduced Planck constant)**.**

$$\hbar = \frac{h}{2\pi}. \tag{8}$$

It carries units of angular momentum, so $[\hbar] = [L_z] = \frac{kg \cdot m^2}{s}$.

Writing $\vec{S} = \hbar(\vec{S}/\hbar)$ with $\vec{S}/\hbar$ dimensionless, the $\hbar$ joins the gyromagnetic ratio. The combination $q\hbar/2mc$ then carries the units and sets the natural size of the moment at a given mass scale.

Definition 9 (Bohr magneton)**.** *The Bohr magneton is the natural unit of magnetic moment for an electron,*

$$\mu_B = \frac{|e|\hbar}{2m_e c}. \tag{9}$$

Definition 10 (Nuclear magneton)**.** *The nuclear magneton is the same combination with the proton mass, and is the natural unit for nucleons,*

$$\mu_N = \frac{|e|\hbar}{2m_p c}. \tag{10}$$

Since $m_p/m_e \approx 1836$, we have $\mu_N \approx \mu_B/1836$.

Definition 11 (Nucleons)**.** *The subatomic particles that make up the nucleus of an atom, specifically protons and neutrons.*

The electron moment is then written as

$$\vec{\mu}_e = g_e \, \mu_B \, \frac{\vec{S}}{\hbar}, \tag{11}$$

and similarly for the proton and neutron with μ_N in place of μ_B.

2.2.3 Values for the electron, proton and neutron

Particle	Charge q	Mass m	g-factor	Gyromagnetic ratio γ
Electron	$-\|e\|$	m_e	$g_e \approx 2$	$g_e\mu_B/\hbar$
Proton	$+\|e\|$	m_p	$g_p \approx 5.59$	$g_p\mu_N/\hbar$
Neutron	0	m_p	$g_n \approx -3.83$	$g_n\mu_N/\hbar$

Table 1: Charges, masses, g-factors and gyromagnetic ratios.

Remark 3. *The classical derivation gives $\mu = (q/2mc)L_z$, which vanishes when $q = 0$. The neutron is neutral yet has a nonzero moment, so its $\vec{\mu}$ cannot come from a single circulating point charge. This is a first sign that the classical picture fails.*

Remark 4. *Since $m_p \gg m_e$, we get $\mu_N \ll \mu_B$ and therefore $|\vec{\mu}_p| \ll |\vec{\mu}_e|$. Stern-Gerlach deflection of nuclei is much weaker than for electrons.*

2.3 The Stern-Gerlach apparatus

Definition 12 (Stern-Gerlach apparatus)**.** *A device that measures one component of a particle's magnetic moment. A beam of neutral atoms passes through a magnet shaped to produce a strong field gradient along a chosen axis, here z. The gradient deflects each atom by an amount proportional to μ_z, and a screen records where it lands. Since $\vec{\mu} = \gamma\vec{S}$, the position on the screen is a direct measurement of S_z.*

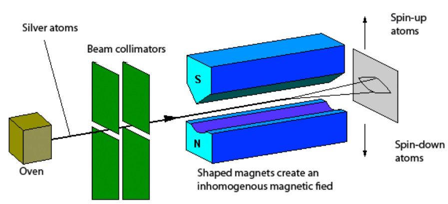

Figure 1: Illustration of the Stern-Gerlach apparatus.

2.3.1 Force in an inhomogeneous field

A magnetic moment in a field has potential energy

$$U = -\vec{\mu} \cdot \vec{B}, \tag{12}$$

so the force on it is

$$\vec{F} = -\vec{\nabla} U = \vec{\nabla}(\vec{\mu} \cdot \vec{B}), \qquad F_z = -\frac{\partial U}{\partial z} = \mu_z \frac{\partial B_z}{\partial z}. \tag{13}$$

Definition 13 (Deflection)**.** *The transverse displacement of an atom on the screen, produced by the force F_z acting over the length of the magnet. It is proportional to μ_z, so the pattern on the screen is a record of the moments in the beam.*

Remark 5. *A uniform field gives no force, only a torque. The apparatus needs a field* gradient *to deflect anything.*

2.3.2 Silver atoms

Silver is used because its net orbital angular momentum vanishes, $\vec{L} = 0$, while $\vec{S} \neq 0$. The entire magnetic moment comes from a single unpaired valence electron, so the atom behaves as one electron spin. For the electron,

$$\mu_z = -\frac{g_e|e|}{2m_e c} S_z, \tag{14}$$

and therefore

$$F_z = \mu_z \frac{\partial B_z}{\partial z} = -g_e \frac{|e|}{2m_e c} S_z \frac{\partial B_z}{\partial z} = g_e \frac{|e|}{2m_e c} \left| \frac{\partial B_z}{\partial z} \right| S_z, \tag{15}$$

where the last equality assumes $\partial B_z / \partial z < 0$.

Result 1.

$$F_z \propto S_z. \tag{16}$$

The deflection of an atom measures the z component of its spin directly.

2.3.3 The classical prediction

Spin is a vector,

$$\vec{S} = (S_x, S_y, S_z)^T, \qquad S_z = |\vec{S}| \cos\theta. \tag{17}$$

Remark 6. *The spins leaving the oven point in random directions, so θ takes every value in $[0, \pi]$ across the beam and $S_z = |\vec{S}| \cos\theta$ takes every value in $[-|\vec{S}|, |\vec{S}|]$. Since $F_z \propto S_z$, each atom is deflected differently and the screen should show a continuous smear. It shows two spots.*

2.4 Quantization of S_z

Result 2 (Quantization of spin)**.** *The electron spin angular momentum is quantized:*

$$S_z = \pm\frac{\hbar}{2}. \tag{18}$$

The beam splits into two spots.

2.4.1 Ket notation

Definition 14 (Ket notation)**.** *A ket $|\cdot\rangle$ labels a quantum state as a vector in a vector space. For electron spin the two measurement outcomes give the basis states*

$$|\uparrow\rangle \text{ for } S_z = +\frac{\hbar}{2}, \qquad |\downarrow\rangle \text{ for } S_z = -\frac{\hbar}{2}.$$

The space is two-dimensional, so any spin state is a linear combination of the two.

Remark 7. *From here on we follow Townsend and write $|+\hat{z}\rangle \equiv |\uparrow\rangle$ and $|-\hat{z}\rangle \equiv |\downarrow\rangle$. The same labels work for any axis: $|\pm\hat{x}\rangle$ are the states with $S_x = \pm\hbar/2$.*

2.5 Sequential Stern-Gerlach experiments

We write SGz for a device whose gradient points along z, and SGx for the same device rotated by 90° about the beam axis. Each device sends atoms into two beams. Blocking one of them turns the device into a filter: every atom that gets through has a known spin component.

A beam straight from the oven always splits 50/50 in any SG device, whatever its orientation. The interesting results come from sending a filtered beam into a second (or third) device.

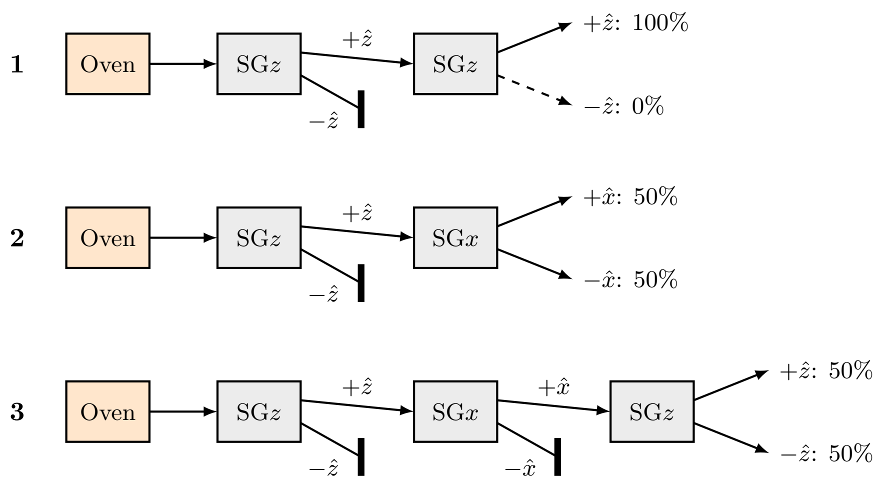

Figure 2: Sequential Stern-Gerlach experiments. Thick bars block a beam. Percentages are the fractions of atoms entering the last device.

2.5.1 Experiment 1: SGz then SGz

Filter the $+\hat{z}$ beam and send it into a second SGz. Every atom comes out $+\hat{z}$, none come out $-\hat{z}$. The measurement is repeatable: once an atom is found with $S_z = +\hbar/2$, measuring S_z again gives the same answer. The atom leaves the first device in the state $|+\hat{z}\rangle$.

2.5.2 Experiment 2: SGz then SGx

Filter the $+\hat{z}$ beam and send it into an SGx. Half the atoms come out $+\hat{x}$ and half come out $-\hat{x}$. Knowing $S_z = +\hbar/2$ tells us nothing about S_x.

2.5.3 Experiment 3: SGz, SGx, then SGz

Filter $+\hat{z}$, then filter $+\hat{x}$, then measure S_z again. The beam splits 50/50 into $+\hat{z}$ and $-\hat{z}$, even though every atom was filtered to $+\hat{z}$ at the start. The S_x measurement wiped out the earlier S_z information.

Result 3. *An atom with a definite value of S_x does not have a definite value of S_z. Measuring S_x destroys what we knew about S_z, so no state has definite values of both at once.*

3 Quantum States as Vectors

September 11, 2026

3.1 Superposition

Experiment 3 shows that an atom in the state $|+\hat{x}\rangle$ does not have a definite $S_z = \pm\hbar/2$. It is neither $|+\hat{z}\rangle$ nor $|-\hat{z}\rangle$, but a combination of both.

Definition 15 (Superposition)**.** *A general spin state is a linear combination of the basis kets,*

$$|\psi\rangle = c_+ |+\hat{z}\rangle + c_- |-\hat{z}\rangle, \qquad c_\pm \in \mathbb{C}. \tag{19}$$

The pair $\{|+\hat{z}\rangle, |-\hat{z}\rangle\}$ is a basis, and the coefficients $c_\pm$ are complex numbers.

Remark 8. *This works exactly like an ordinary vector written in components,*

$$\vec{E} = E_x\hat{x} + E_y\hat{y} + E_z\hat{z}, \tag{20}$$

where the basis is orthonormal:

$$\hat{x}\cdot\hat{x} = \hat{y}\cdot\hat{y} = \hat{z}\cdot\hat{z} = 1, \qquad \hat{x}\cdot\hat{y} = \hat{x}\cdot\hat{z} = \hat{y}\cdot\hat{z} = 0. \tag{21}$$

A component is picked out with a dot product, $E_x = \hat{x}\cdot\vec{E}$. Kets do the same thing, with the inner product playing the role of the dot product.

3.2 Bras and the inner product

Definition 16 (Bra)**.** *Every ket $|\psi\rangle$ has a matching bra $\langle\psi|$. Putting a bra in front of a ket gives a complex number, the inner product*

$$\langle\phi|\,|\psi\rangle \equiv \langle\phi|\psi\rangle \in \mathbb{C}. \tag{22}$$

Definition 17 (Orthonormal basis)**.** *The basis kets are orthonormal:*

$$\langle+\hat{z}|+\hat{z}\rangle = \langle-\hat{z}|-\hat{z}\rangle = 1, \qquad \langle+\hat{z}|-\hat{z}\rangle = \langle-\hat{z}|+\hat{z}\rangle = 0. \tag{23}$$

3.3 Probability amplitudes

To pull out c_+, act on $|\psi\rangle$ with $\langle+\hat{z}|$, just like dotting $\vec{E}$ with $\hat{x}$:

$$\langle+\hat{z}|\psi\rangle = c_+ \underbrace{\langle+\hat{z}|+\hat{z}\rangle}_{=1} + c_- \underbrace{\langle+\hat{z}|-\hat{z}\rangle}_{=0} = c_+. \tag{24}$$

The same steps with $\langle-\hat{z}|$ give c_-. So

$$c_\pm = \langle\pm\hat{z}|\psi\rangle. \tag{25}$$

Definition 18 (Probability amplitude)**.** *The coefficient* $c_\pm = \langle \pm\hat{z}|\psi\rangle$ *is the probability amplitude for measuring* $S_z = \pm\hbar/2$*. The probability itself is the modulus squared,*

$$P\big(S_z = \pm\tfrac{\hbar}{2}\big) = |c_\pm|^2 = |\langle \pm\hat{z}|\psi\rangle|^2 \,. \tag{26}$$

Principle 1 (Born rule)**.** *The probability that a system in the state* $|\psi\rangle$ *is found in the state* $|\phi\rangle$ *is*

$$P(\phi\,|\,\psi) = |\langle\phi|\psi\rangle|^2 \in [0,1]. \tag{27}$$

3.4 Normalization

A system in $|\psi\rangle$ is found in $|\psi\rangle$ with certainty, so

$$P(\psi\,|\,\psi) = |\langle\psi|\psi\rangle|^2 = 1. \tag{28}$$

This fixes what the bra looks like. Write it with unknown coefficients, $\langle\psi| = c'_+ \langle +\hat{z}| + c'_- \langle -\hat{z}|$. Then

$$\langle\psi|\psi\rangle = \big(c'_+ \langle +\hat{z}| + c'_- \langle -\hat{z}|\big)\left(c_+ |+\hat{z}\rangle + c_- |-\hat{z}\rangle\right) = c'_+ c_+ + c'_- c_-, \tag{29}$$

since the cross terms vanish by orthonormality. The two outcomes $S_z = \pm\hbar/2$ must have probabilities $|c_+|^2$ and $|c_-|^2$ that add to 1. For $\langle\psi|\psi\rangle$ to be that sum, each product $c'_\pm c_\pm$ must be real and equal to $|c_\pm|^2$. This forces $c'_\pm = c^*_\pm$.

Result 4. *Going from a ket to its bra, conjugate the coefficients:*

$$\langle\psi| = c^*_+ \langle +\hat{z}| + c^*_- \langle -\hat{z}| \,. \tag{30}$$

In particular $\langle\psi|\pm\hat{z}\rangle = c^*_\pm = \langle\pm\hat{z}|\psi\rangle^*$*. More generally,* $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$*.*

Definition 19 (Normalization condition)**.**

$$\langle\psi|\psi\rangle = \langle\psi|+\hat{z}\rangle\langle+\hat{z}|\psi\rangle + \langle\psi|-\hat{z}\rangle\langle-\hat{z}|\psi\rangle = |c_+|^2 + |c_-|^2 = 1. \tag{31}$$

In words, $P(S_z = +\hbar/2) + P(S_z = -\hbar/2) = 1$*: the atom is always found in one of the two beams.*

Example. *Take*

$$|\psi\rangle = \frac{1}{\sqrt{3}}|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}|-\hat{z}\rangle \,.$$

The amplitude for $S_z = +\hbar/2$ *is*

$$\langle+\hat{z}|\psi\rangle = \frac{1}{\sqrt{3}}\underbrace{\langle+\hat{z}|+\hat{z}\rangle}_{=1} + i\sqrt{\frac{2}{3}}\underbrace{\langle+\hat{z}|-\hat{z}\rangle}_{=0} = \frac{1}{\sqrt{3}},$$

so $P(S_z = +\hbar/2) = 1/3$*. In the same way* $\langle-\hat{z}|\psi\rangle = i\sqrt{2/3}$*, so* $P(S_z = -\hbar/2) = |i|^2 \cdot 2/3 = 2/3$*. The two add to 1, so the state is normalized. The factor* i *changes nothing here since* $|i| = 1$*.*

3.5 The state $|+\hat{x}\rangle$

In Experiment 3, atoms in $|+\hat{x}\rangle$ enter an SGz and split 50/50:

$$P\big(S_z = \pm\tfrac{\hbar}{2}\big) = |\langle \pm\hat{z}| + \hat{x}\rangle|^2 = \frac{1}{2}. \tag{32}$$

So both amplitudes have modulus $1/\sqrt{2}$, and each can carry its own phase:

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} e^{i\delta_+} |+\hat{z}\rangle + \frac{1}{\sqrt{2}} e^{i\delta_-} |-\hat{z}\rangle = \frac{e^{i\delta_+}}{\sqrt{2}} \big(|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle\big), \qquad \delta = \delta_- - \delta_+, \tag{33}$$

with $\delta_\pm \in \mathbb{R}$.

3.6 Global phase

Take two kets that differ only by an overall phase, $|\psi\rangle = e^{i\alpha} |\psi'\rangle$ with $\alpha \in \mathbb{R}$. For any state $|\phi\rangle$,

$$P(\phi \,|\, \psi) = |\langle \phi|\psi\rangle|^2 = \left|e^{i\alpha} \langle \phi|\psi'\rangle\right|^2 = |\langle \phi|\psi'\rangle|^2 = P(\phi \,|\, \psi'), \tag{34}$$

since $|e^{i\alpha}| = 1$.

Result 5. *$|\psi\rangle$ and $e^{i\alpha} |\psi\rangle$ give the same probabilities for every possible measurement, so they describe the same physical state. An overall (global) phase has no physical meaning and can be dropped.*

Dropping the global phase $e^{i\delta_+}$ from $|+\hat{x}\rangle$ leaves

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} \big(|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle\big). \tag{35}$$

Remark 9. *The relative phase δ is different: it sits between the two terms, so it cannot be pulled out front and dropped. Experiment 3 alone does not fix it. It is pinned down by combining more measurements (for example along y) with a phase convention, and the standard choice for $|+\hat{x}\rangle$ is $\delta = 0$.*

A Appendix

B Useful Links

Stern-Gerlach experiment: https://www.youtube.com/shorts/QG7kJXxN1Kk